

Multi-Class Text Classification: A Comparison of Word Representations and ML/NN Models

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Abstract—This project used the dataset of online questions into different categories. We explored the dataset, cleaned the text dataset, and experimented with different machine learning and deep learning models using different text representations like Bag of Words, TF-IDF, GloVe, and Skip-gram embeddings. Our goal was to identify the most effective approach for this question classification task.

I. INTRODUCTION

Online platforms often feature large volumes of user-generated questions. Accurately categorizing these questions is crucial for efficient content management, improved search functionality and better user experience. This project aimed to build and evaluate different models to see the performance based on their text content. The dataset used contains a large collection of questions and their corresponding categories.

II. METHODOLOGY

This section outlines the experimental methodology employed in the project, including dataset handling, exploratory data analysis (EDA), preprocessing, feature extraction, model architectures, hyperparameter tuning, and evaluation strategy. The workflow was designed to satisfy the course requirement of conducting 22 experiments across multiple word representation techniques and models.

A. Dataset and Experimental Splits

The assigned dataset was provided with a predefined 80/20 train-test split. To further support model selection and hyperparameter tuning, the training set was partitioned into training and validation subsets using an additional 80/20 split. All random processes were seeded to ensure reproducibility. The implementation was carried out in Python, leveraging scikit-learn for machine learning models, TensorFlow/Keras for neural architectures, Gensim for word embeddings, and NLTK/spaCy for natural language preprocessing.

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B. Exploratory Data Analysis

Exploratory data analysis was conducted to assess the dataset and guide preprocessing. Class distribution was inspected to identify imbalance, and token-length statistics were used to define a maximum sequence length for neural models. Noisy or empty samples, including those containing artifacts such as URLs or HTML tags, were removed to improve data quality.

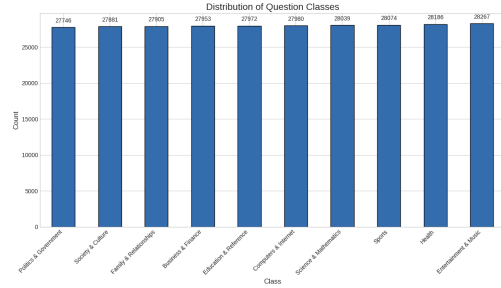


Fig. 1. Distribution of Question Classes

C. Preprocessing

The preprocessing pipeline normalized text and reduced noise through lowercasing, whitespace removal, and elimination of URLs, emails, numbers, and excessive punctuation. Tokenization was performed using `CountVectorizer`/`TfidfVectorizer` for machine learning models and the `Keras Tokenizer` for neural models. Stopword removal was applied to BoW and TF-IDF features, while lemmatization with the `NLTK WordNet` lemmatizer ensured consistent word forms. The exact preprocessing steps varied slightly across experiments based on validation results.

D. Word Representations

Four word representation methods were used. Bag-of-Words features were built with `CountVectorizer`, limited to 10,000 tokens (mainly unigrams, with bigrams in ablations).

TABLE I
PERFORMANCE OF MACHINE LEARNING MODELS

Model + Representation	Accuracy	F1-Macro
Random Forest (BoW)	0.5239	0.5273
Random Forest (TF-IDF)	0.5252	0.5279
Logistic Regression (BoW)	0.6560	0.6511
Logistic Regression (TF-IDF)	0.6700	0.6658
Naive Bayes (BoW)	0.6461	0.6427
Naive Bayes (TF-IDF)	0.6519	0.6480

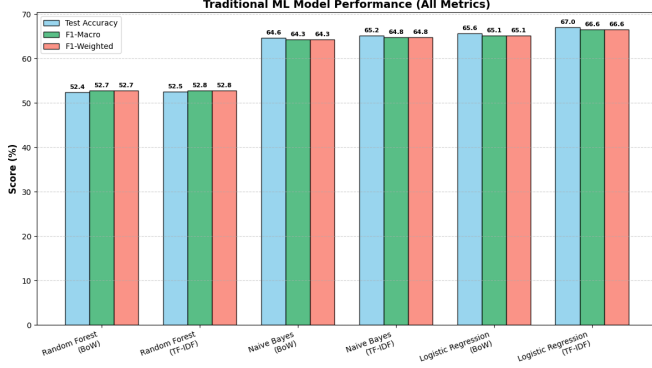


Fig. 3. Comparison of Machine Learning Models across BoW and TF-IDF representations.

B. Performance of Neural Models with Sparse Features

Deep Neural Networks (DNNs) were also trained on sparse representations, namely BoW and TF-IDF. As shown in Table II, these models underperformed relative to Logistic Regression. The results suggest that sparse, high-dimensional representations are better suited for linear classifiers rather than dense neural networks, which benefit more from continuous word embeddings.

TABLE II
PERFORMANCE OF DNN WITH SPARSE FEATURES

Model + Representation	Accuracy	F1-Macro
DNN (BoW)	0.6221	0.6154
DNN (TF-IDF)	0.6481	0.6419

C. Performance of Neural Models with Dense Embeddings

Neural models were further evaluated with pre-trained GloVe and Skip-gram embeddings. Results for GloVe embeddings are presented in Table III and visualized in Figure 4. SimpleRNN-based models performed poorly, with macro F1 values as low as 0.08, demonstrating that shallow recurrent architectures were unable to capture long-range dependencies. In contrast, GRU, LSTM, and their bidirectional variants performed substantially better, with the Bidirectional GRU achieving the strongest performance within GloVe-based models, reaching 67.3% accuracy and a macro F1 of 0.6674.

Skip-gram embeddings consistently outperformed GloVe across nearly all neural architectures. As shown in Table IV and Figure 5, recurrent architectures such as GRU and LSTM showed significant improvements when using Skip-gram. The

TABLE III
NEURAL MODEL PERFORMANCE WITH GLOVE EMBEDDINGS

Model	Accuracy	F1-Macro
DNN	0.5750	0.5645
SimpleRNN	0.1591	0.0820
GRU	0.6663	0.6582
LSTM	0.6624	0.6540
BiRNN	0.4897	0.4732
BiGRU	0.6732	0.6674
BiLSTM	0.6622	0.6549

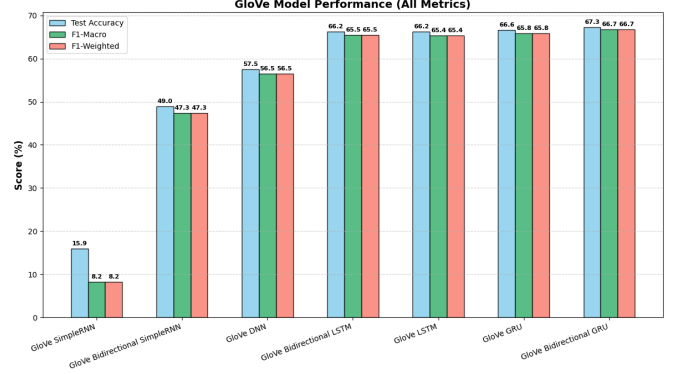


Fig. 4. Performance of neural models trained with GloVe embeddings across all metrics.

Bidirectional GRU with Skip-gram embeddings achieved the best overall result among all neural networks, with a test accuracy of 69.0% and macro F1 of 0.6823.

TABLE IV
NEURAL MODEL PERFORMANCE WITH SKIP-GRAM EMBEDDINGS

Model	Accuracy	F1-Macro
DNN	0.6828	0.6760
SimpleRNN	0.2036	0.1289
GRU	0.6876	0.6829
LSTM	0.6833	0.6793
BiRNN	0.5754	0.5642
BiGRU	0.6900	0.6823
BiLSTM	0.6842	0.6792

D. Comparative Analysis Across All Models

The results of all twenty-two models are compared in Figure 6, which ranks the models based on test accuracy. Logistic Regression with TF-IDF emerged as the strongest traditional ML model, while the Bidirectional GRU with Skip-gram embeddings was the best-performing neural model. At the opposite end of the spectrum, SimpleRNN and Bidirectional RNN models trained with GloVe embeddings consistently showed the weakest results. These findings confirm that sparse representations such as TF-IDF are particularly effective for linear models, while dense embeddings such as Skip-gram enable advanced recurrent architectures to capture contextual dependencies more effectively.

E. Neural Network Architecture-Level Comparison

To further analyze the influence of architecture independent of embedding choice, scores were averaged across embeddings

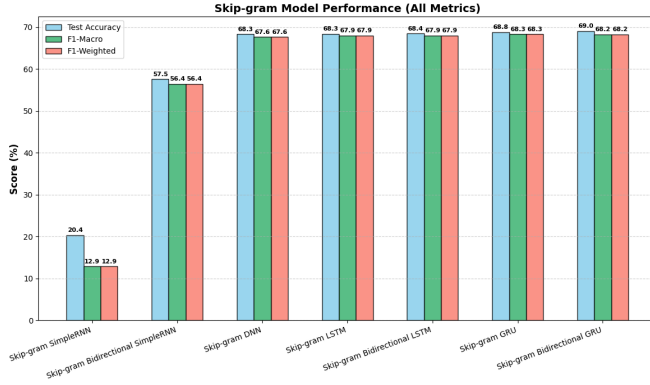


Fig. 5. Performance of neural models trained with Skip-gram embeddings across all metrics.

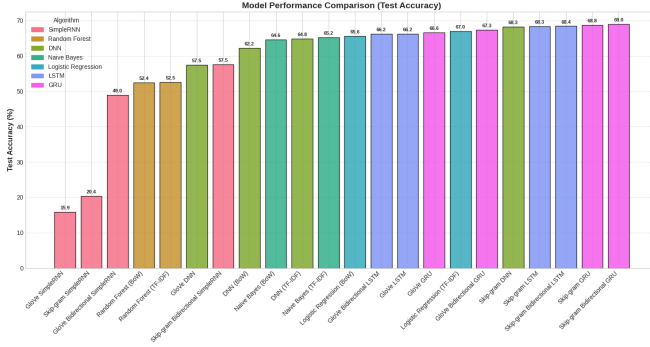


Fig. 6. Overall performance comparison of all models (ML + NN) ranked by test accuracy.

for each neural architecture type. The comparison in Figure ?? shows that GRU and LSTM architectures consistently achieved the highest overall performance, followed closely by DNN, while SimpleRNN remained the weakest model family. This reinforces the observation that gating mechanisms, as in GRU and LSTM, play a crucial role in handling sequential dependencies in textual data, particularly when coupled with effective embeddings. Moreover, the stability of results across embeddings indicates that architectural design exerts a stronger influence on performance than representation choice alone. This suggests that selecting robust recurrent units is a critical factor for achieving generalizable improvements in text classification.

F. Best and Worst Performing Models

To provide a holistic view of model behavior, both visual and tabular representations were employed to compare the performance of the twenty-two experiments. The analysis focused on identifying the strongest and weakest performers among both traditional machine learning models and neural network architectures.

1) *Traditional Machine Learning Models*: Among machine learning approaches, Logistic Regression with TF-IDF features achieved the highest performance, reaching 67.0% accuracy and 0.6658 macro-F1. In contrast, Random Forest trained

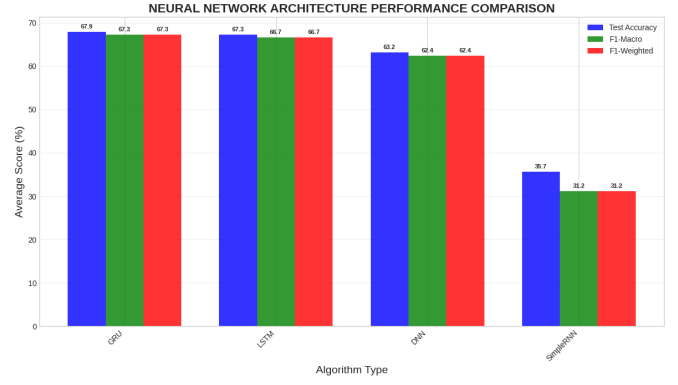


Fig. 7. Comparison of neural network architectures averaged across embeddings. GRU and LSTM achieved the highest overall performance, while SimpleRNN remained the weakest.

TABLE V
PERFORMANCE OF NEURAL NETWORK ARCHITECTURES (ACCURACY, F1-MACRO, F1-WEIGHTED)

Model	Accuracy (%)	F1-Macro (%)	F1-Weighted (%)
DNN (BoW)	62.21	61.54	61.54
DNN (TF-IDF)	64.81	64.19	64.19
DNN (GloVe)	57.50	56.45	56.45
DNN (Skip-gram)	68.28	67.60	67.60
SimpleRNN (GloVe)	15.91	8.20	8.20
SimpleRNN (Skip-gram)	20.36	12.89	12.89
BiRNN (GloVe)	48.97	47.32	47.32
BiRNN (Skip-gram)	57.54	56.42	56.42
GRU (GloVe)	66.63	65.82	65.82
GRU (Skip-gram)	68.76	68.29	68.29
BiGRU (GloVe)	67.32	66.74	66.74
BiGRU (Skip-gram)	69.00	68.23	68.23
LSTM (GloVe)	66.24	65.40	65.40
LSTM (Skip-gram)	68.33	67.93	67.93
BiLSTM (GloVe)	66.22	65.49	65.49
BiLSTM (Skip-gram)	68.42	67.92	67.92

on Bag-of-Words features yielded the weakest performance, with only 52.4% accuracy and 0.5273 macro-F1. Table VI summarizes these results, while Figure 8 provides a visual comparison. The confusion matrices (Figures 9 and 10) further reveal how Logistic Regression demonstrated more balanced predictions across categories, whereas Random Forest exhibited frequent misclassifications, particularly in semantically overlapping classes.

TABLE VI
BEST AND WORST PERFORMING MACHINE LEARNING MODELS

Model	Accuracy (%)	F1-Macro	F1-Weighted
Logistic Regression (TF-IDF)	67.0	0.6658	0.6660
Random Forest (BoW)	52.4	0.5273	0.5273

2) *Neural Network Models*: For neural networks, the Bidirectional GRU trained with Skip-gram embeddings emerged as the best performer, achieving 69.0% accuracy and 0.6823 macro-F1. Conversely, the SimpleRNN trained with GloVe embeddings was the weakest, producing only 15.9% accuracy and a macro-F1 score of 0.0820. Table VII presents

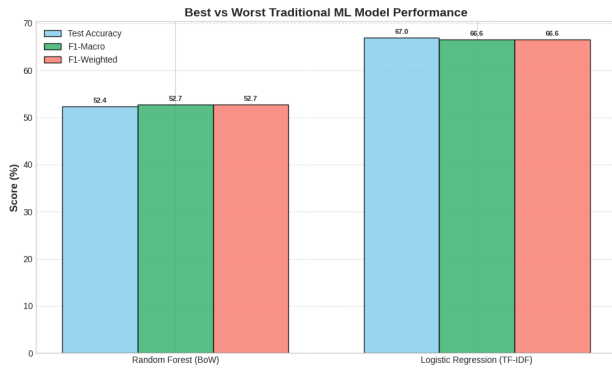


Fig. 8. Performance comparison of best and worst ML models.

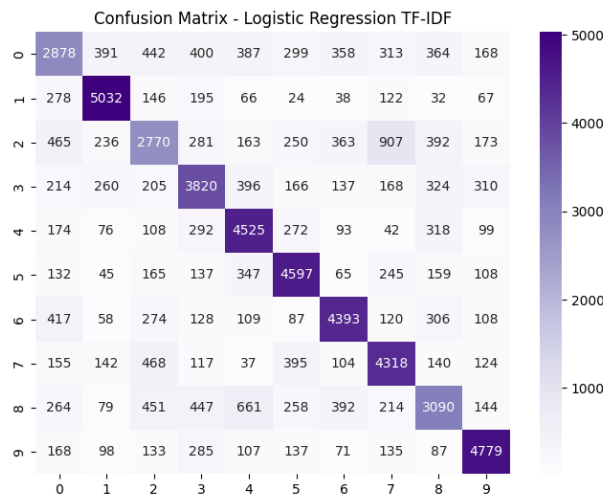


Fig. 9. Confusion Matrix of Logistic Regression with TF-IDF.

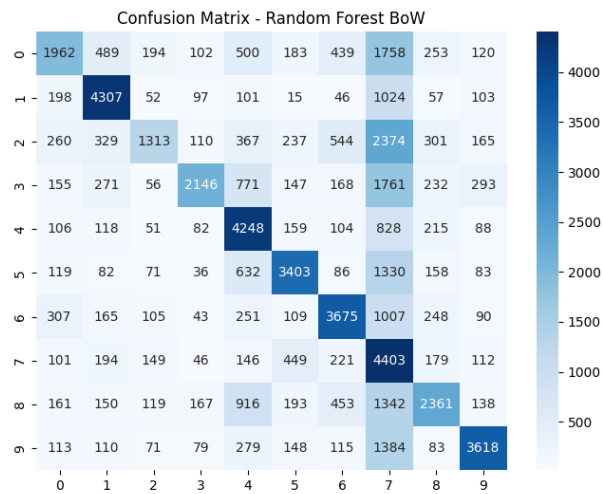


Fig. 10. Confusion Matrix of Random Forest with BoW.

the detailed results, while Figure 11 illustrates the striking contrast. Confusion matrices (Figures 12 and 13) further highlight this difference: the Skip-gram BiGRU produced balanced predictions across all categories, while the GloVe SimpleRNN failed to capture class distinctions, leading to severe misclassification.

TABLE VII
BEST AND WORST PERFORMING NEURAL NETWORK MODELS

Model	Accuracy (%)	F1-Macro	F1-Weighted
Skip-gram BiGRU	69.0	0.6823	0.6820
GloVe SimpleRNN	15.9	0.0820	0.0820

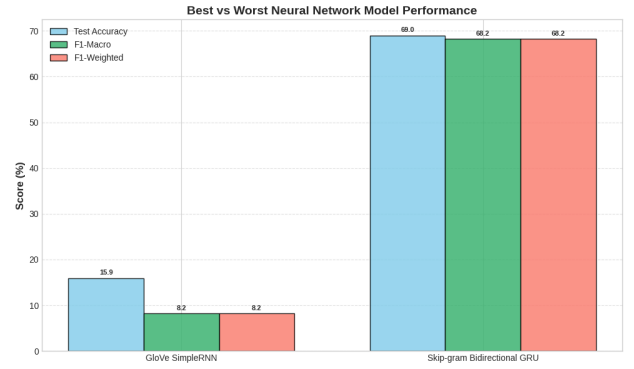


Fig. 11. Performance comparison of best and worst NN models.

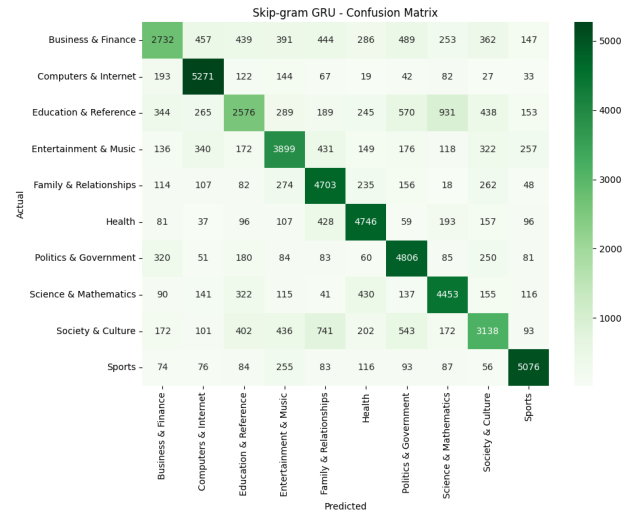


Fig. 12. Confusion Matrix of Skip-gram BiGRU.

3) *Discussion*: Overall, the evidence highlights two important findings. First, sparse representations such as TF-IDF are particularly well-suited for linear classifiers like Logistic Regression, where interpretability and stability are advantageous. Second, dense embeddings such as Skip-gram, when paired with advanced recurrent units like GRU, consistently yield the strongest results by capturing sequential dependencies

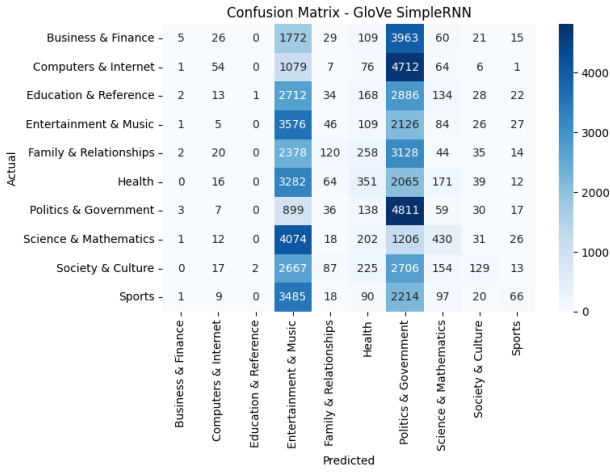


Fig. 13. Confusion Matrix of GloVe SimpleRNN.

and semantic structure. These results confirm that the Skip-gram BiGRU is the most effective configuration for this dataset, whereas SimpleRNN variants, particularly with GloVe embeddings, remain consistently the weakest performers.

G. Comparison of Best-Performing ML and NN Models

To deepen the analysis, a comparison was made between the best-performing machine learning and neural network models. Logistic Regression with TF-IDF achieved 67.0% accuracy and a macro-F1 of 0.6658, benefiting from sparse representations that highlight discriminative terms while remaining simple and interpretable. However, its linear boundaries limited its ability to capture deeper contextual dependencies.

In contrast, the Skip-gram Bidirectional GRU reached 69.0% accuracy and a macro-F1 of 0.6823. By leveraging dense embeddings and bidirectional recurrence, it modeled sequential and semantic relationships more effectively, particularly in overlapping categories such as *Politics* and *Science*. This performance came at the expense of higher computational cost and lower interpretability, illustrating the trade-off between efficiency and contextual modeling.

TABLE VIII
DETAILED COMPARISON OF BEST-PERFORMING ML AND NN MODELS

Aspect	LogReg (TF-IDF)	BiGRU (Skip-gram)
Accuracy (%)	67.0	69.0
F1-Macro	0.6658	0.6823
F1-Weighted	0.6660	0.6820
Input Rep.	Sparse TF-IDF (10k)	Dense Skip-gram (100)
Architecture	Linear softmax	Bi-directional GRU
Context	Independent words	Sequential (past/future)
Strengths	Fast, simple, interpretable	Semantics, seq. modeling
Weaknesses	No long-range deps.	High cost, less interpretable
Training Time	Very low	High (GPU)
Interpretability	High	Low

To further illustrate these distinctions, Figure 14 presents a direct bar-chart comparison of accuracy, macro-F1, and weighted F1 across the two models. Meanwhile, Figure 15 uses a radar chart to contrast them along additional qualitative

dimensions such as training time, interpretability, and context handling. The bar chart emphasizes the slight but consistent performance advantage of BiGRU over Logistic Regression, while the radar chart highlights the trade-off: BiGRU excels in sequential context and semantic modeling, whereas Logistic Regression retains its edge in simplicity, efficiency, and interpretability.

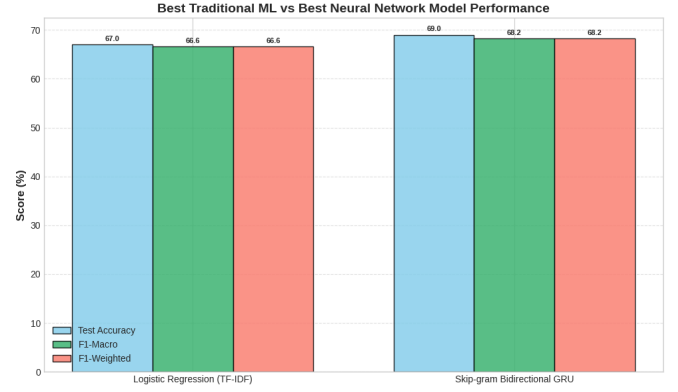


Fig. 14. Best ML (Logistic Regression TF-IDF) vs Best NN (Skip-gram BiGRU) Performance Comparison.

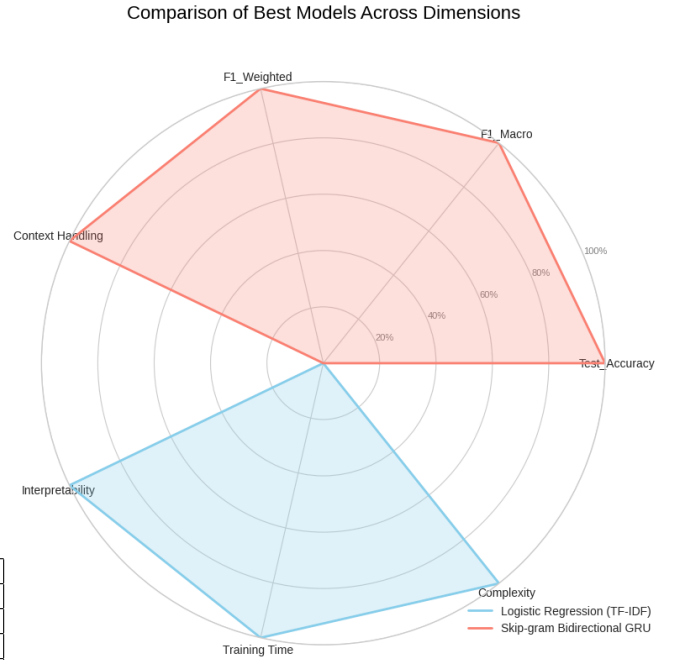


Fig. 15. Radar Chart Comparison Across Quantitative and Qualitative Dimensions.

IV. CONCLUSION

This project presented a comprehensive evaluation of twenty-two models for multi-class question classification, spanning both traditional machine learning approaches with sparse features and neural network architectures with dense embeddings. The results demonstrated that representation

choice plays a decisive role: Logistic Regression with TF-IDF proved to be the most effective machine learning baseline, achieving 67.0% accuracy and a macro-F1 of 0.6658, while Random Forest consistently underperformed. These findings reaffirm the strength of simple, interpretable linear models when paired with discriminative sparse representations.

Neural architectures, on the other hand, benefited substantially from dense embeddings, with the Skip-gram Bidirectional GRU achieving the best overall performance at 69.0% accuracy and 0.6823 macro-F1. The results highlight the importance of gating mechanisms and bidirectional context for capturing sequential dependencies in text. However, these gains came at the cost of increased computational resources and reduced interpretability compared to classical models. Overall, the study suggests that while sparse TF-IDF features remain highly competitive for efficient classification, Skip-gram embeddings combined with advanced recurrent models provide superior semantic modeling. Future work could extend this investigation by exploring transformer-based embeddings, such as BERT, which may further enhance performance through deeper contextualization.

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